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Completed
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Zener diode as voltage Regulation

AIM:-

TO Simulate the line and load regulation operation zener diode

Apparatus Required:

Laptop with proteus software

Theory

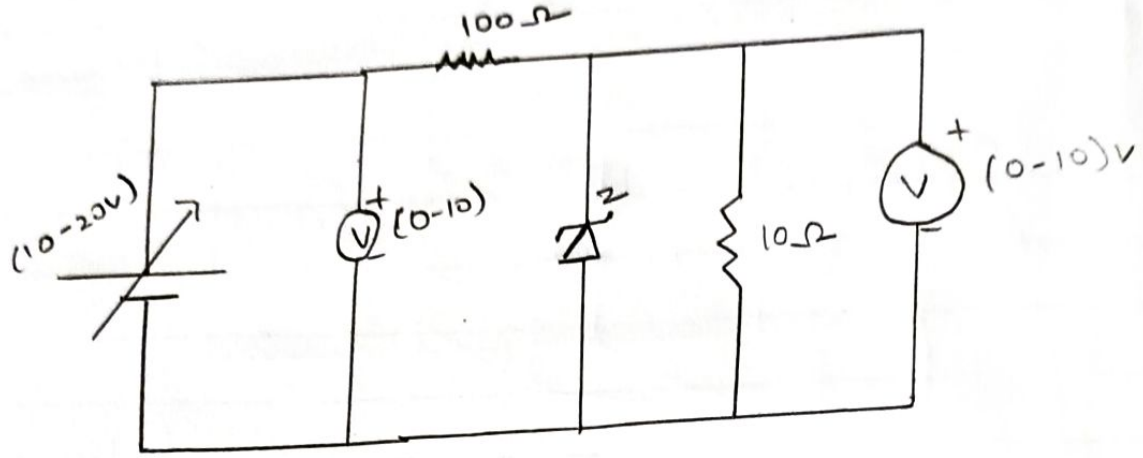
A heavily dopped PN Junction diode is Called zener diode. Due to heavily dopped nature zener diode works under forward bias and reverse bias conditions

Zener diode under reverse bias condition is used as a Voltage regulation. Even if the input voltage changes the constant. This is called line regulation

If the load resistance change the output voltage is constant this is called load regulation

Circuit diagram

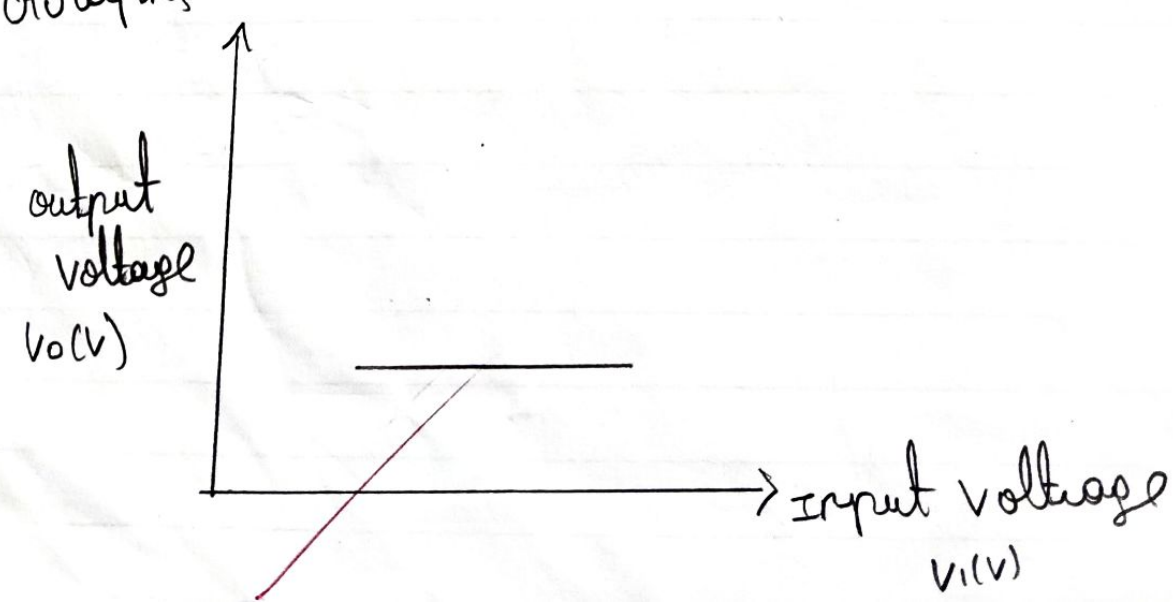
Line regulation



Tabulation

S.NO	Input voltage Vi (V)	output voltage Vo (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

Model Graphs



No. 5

procedure

1) Drag the required component from the protous library

2) Connect the component (required) as per circuit diagram

3) For line regulation vary the input voltage and not the outputed voltage in tabulation. draw the graph between input voltage and output voltage

4) For load regulations vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage

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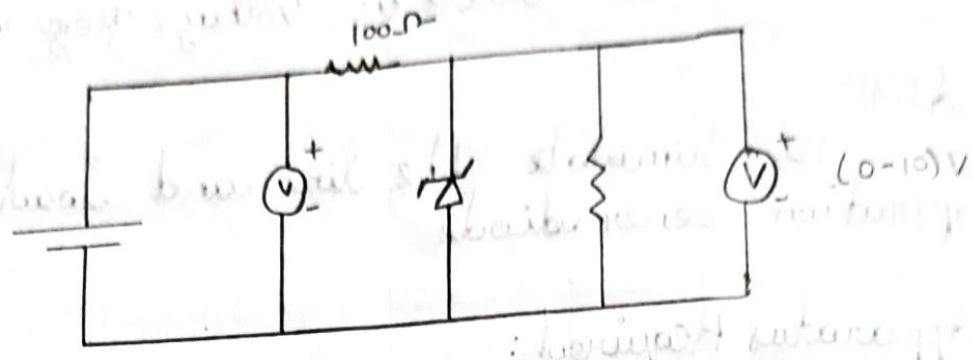
Circuit Diagram

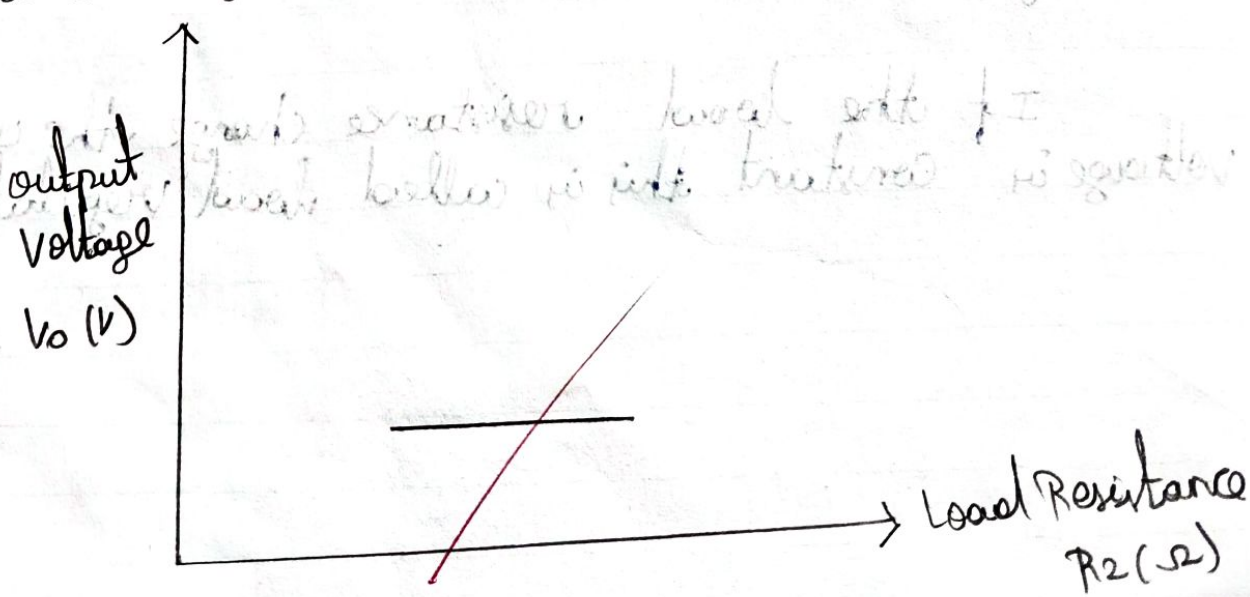
Load Regulation



Tabulation

SNo	Load Resistance $R_2(\Omega)$	output voltage V_o (V)
1	100Ω	+5.19V
2	200Ω	+5.20V
3	300Ω	+5.20V
4	400Ω	+5.20V
5	500Ω	+5.20V
6	600Ω	+5.21V

Model diagram



Expt. No. 5

100

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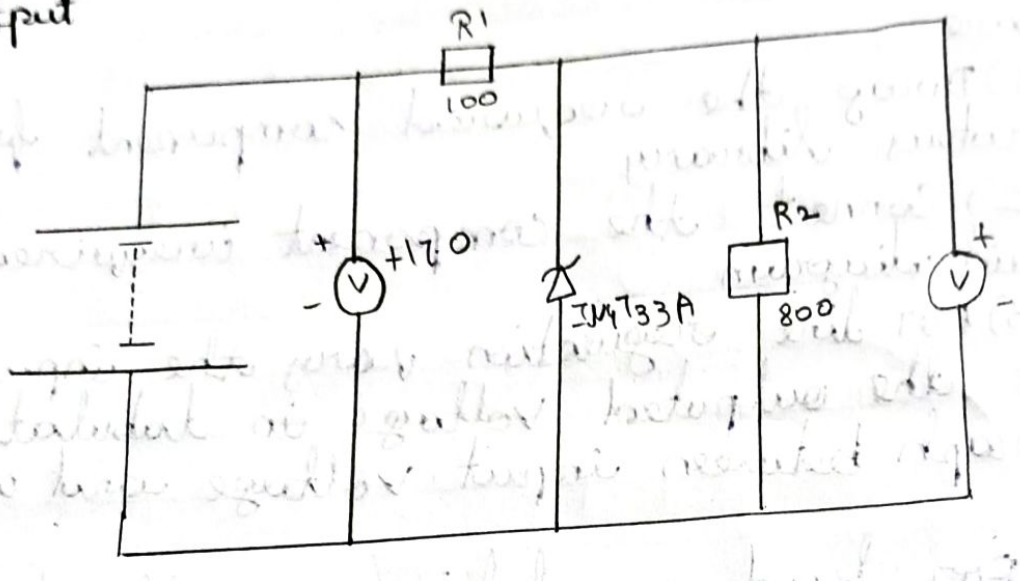
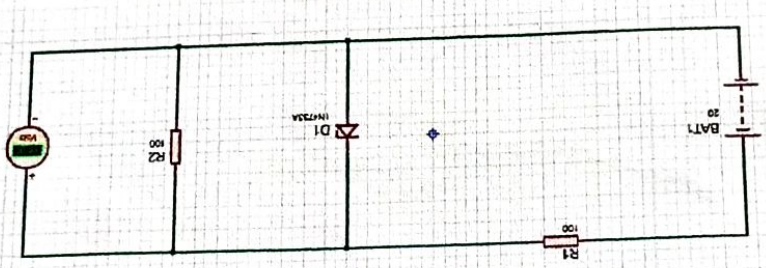
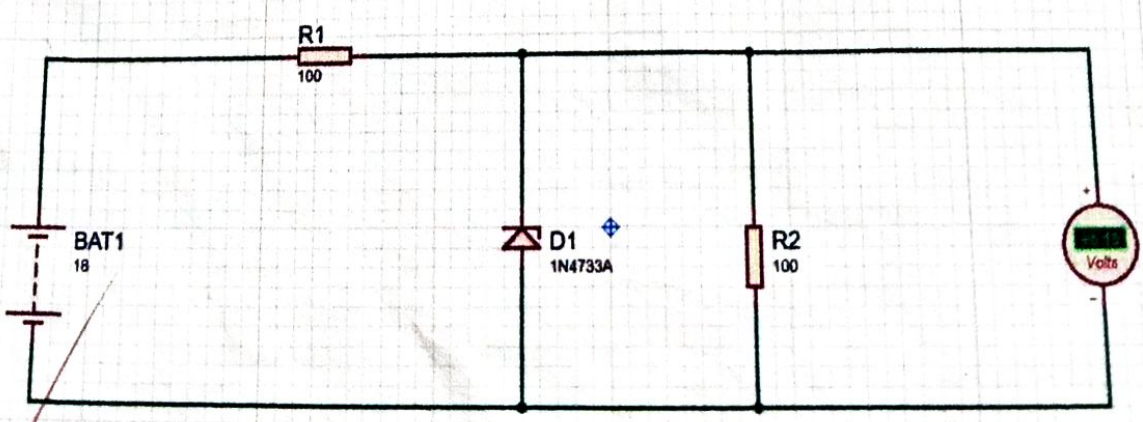
Result:

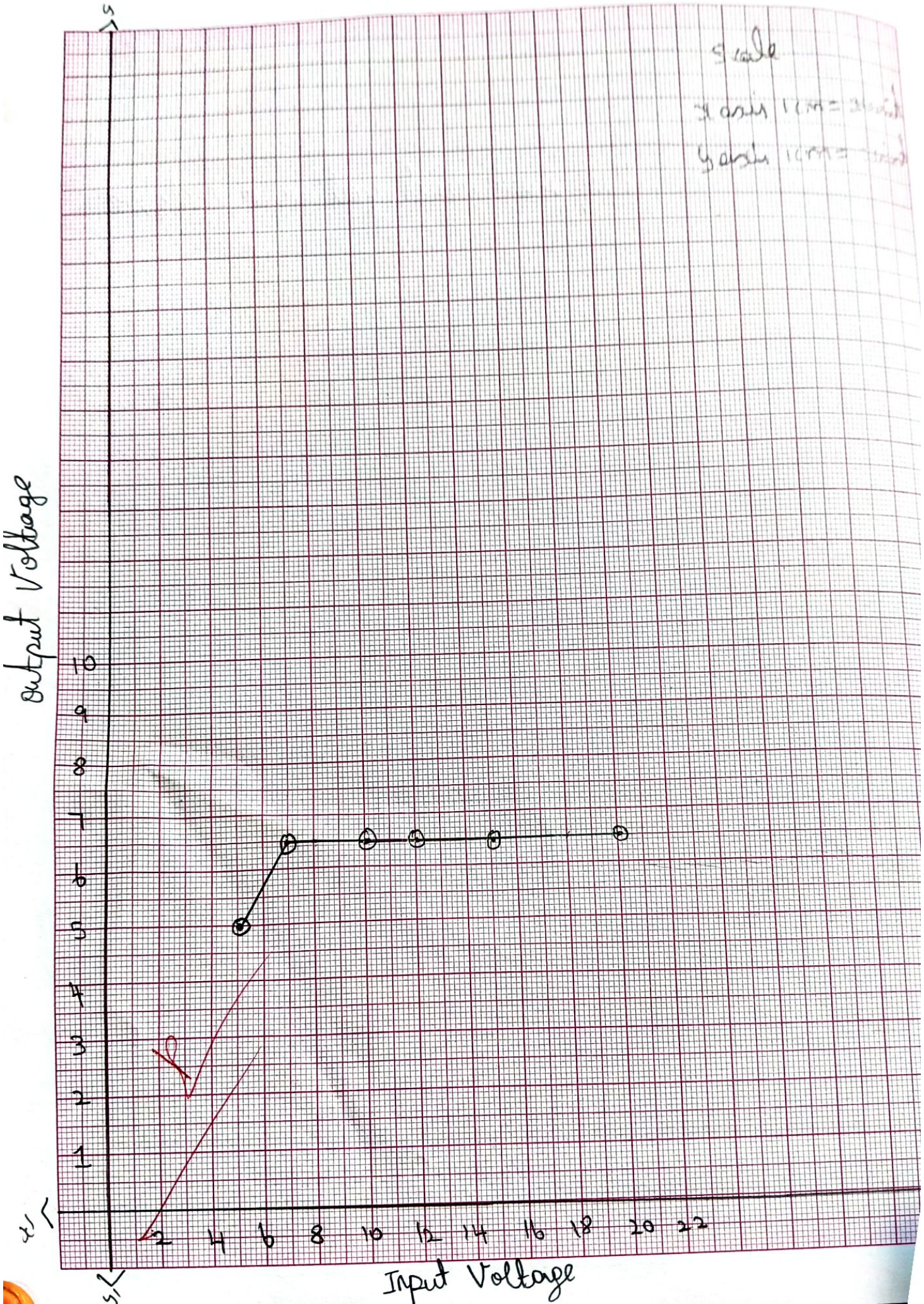
The stimulation of line stand regulation operation of zener diode successfully completed

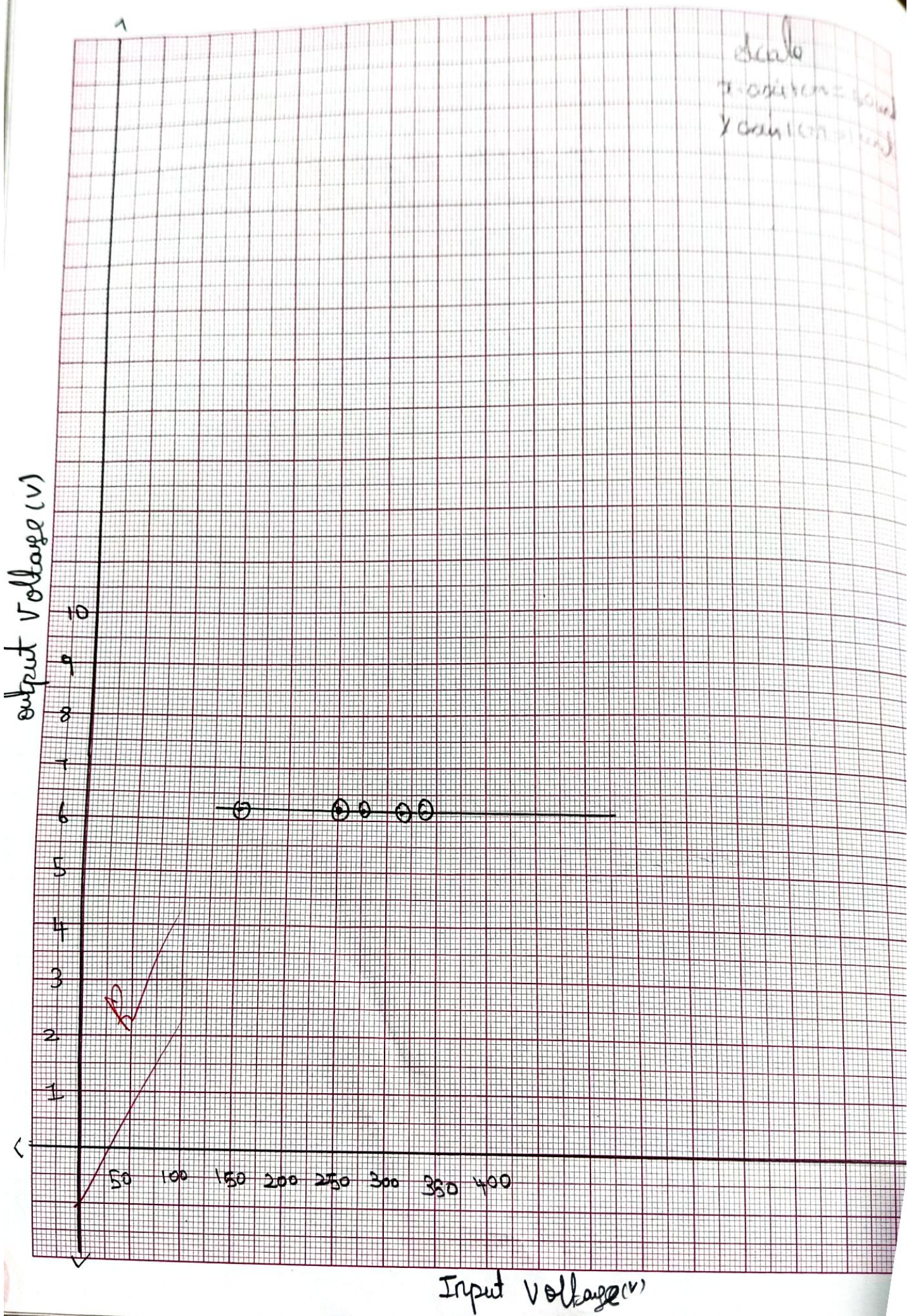
output


R1
100
+17.0
1N4733A
R2
800


R1
100
BAT1
20
D1
1N4733A
R2
100
Volts


R1
100
BAT1
18
D1
1N4733A
R2
100
Volts


Scale
x axis 1cm = 2volt
y axis 1cm = 1volt
Output Voltage
Input Voltage
1
2
3
4
5
6
7
8
9
10
2
4
6
8
10
12
14
16
18
20
22


Scale
X-axis = 50mV
Y-axis = 1V
output voltage (V)
10
9
8
7
6
5
4
3
2
1
50
100
150
200
250
300
350
400
Input voltage (V)